Packet Capture Analysis Strategy

From Protocol Filtering to Detecting Malicious Beacons

By
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Strategic Objectives & Technical Goals

Mastering the Packet-Level Narrative

Core Objectives

- **Strategic Intent Identification**: Shifting from passive observation to active threat hunting.
- **Protocol Mastery**: Hardening defenses against invisible DNS and high-volume traffic.
- **Evidence-Based Logic**: Moving beyond hunches to data-driven investigative process.
- **Operational Discipline**: Establishing a repeatable, rigorous investigative process.

Our Goals

- **Baseline Mastery**: Establishing a clear definition of normal network behavior.
- **Framework Alignment**: Correlating technical artifacts to MITRE ATT&CK techniques.
- **Volume Analysis**: Distinguishing between Elephant exfiltration and Mice heartbeats.
- **Tool Integration**: Utilizing RITA and Zeek to automate long-term detection.

Filtering Methodology

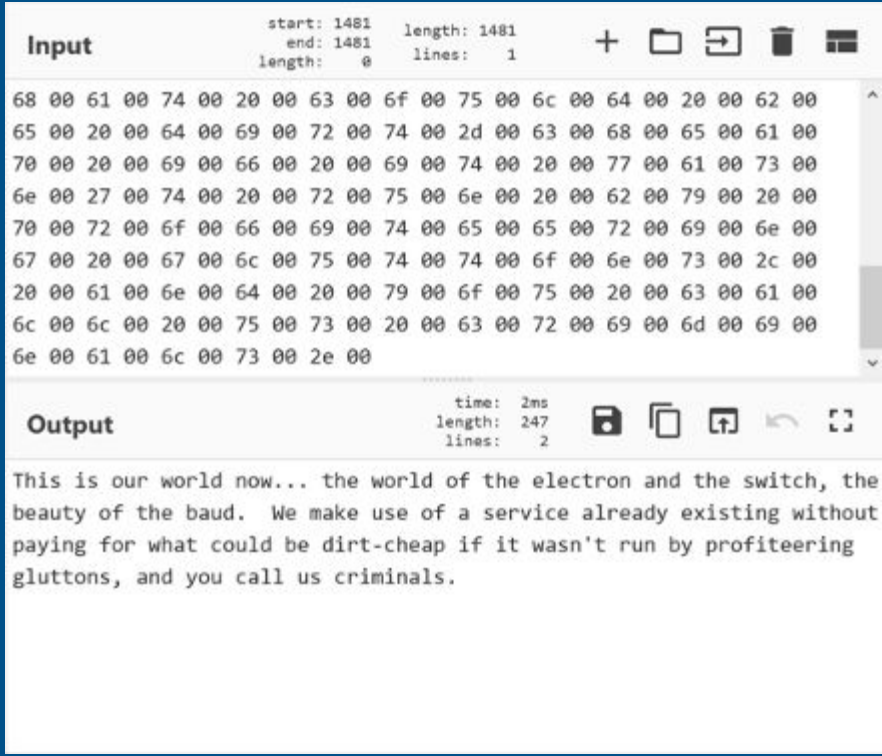
Moving from Packet Collection to Network Storytelling

The Analyst's Mission

- **Efficiency**: Drastically reducing MTTR through systematic pcap navigation.
- **Contextualization**: Turning raw hex data into a coherent narrative of attacker behavior.
- **Precision Analysis**: Ensuring every applied filter serves a specific investigative pivot.

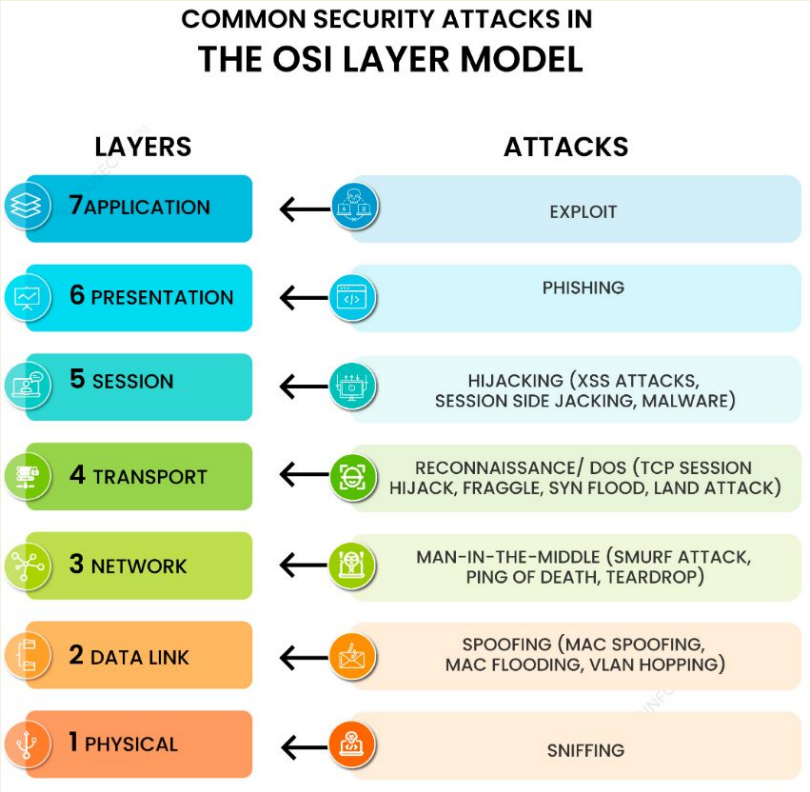
The Strategic Roadmap

- **L2-L7 Scoping**: Using the OSI model to partition the hunt and isolate the attack layer.
- **Boolean Logic**: Applying AND/OR/NOT operators to surgically remove background noise.
- **Session Integrity**: Tracking handshakes and flags toverify successful connections.
- **Artifact Velocity**: Accelerating the discovery of malicious indicators within the firehose.



The Analyst's Lens: The OSI Model

Using the OSI Model as a Technical Map for Isolation



Layered Investigation

- **Transport Health (Layers 2-4)**: Assessing connection integrity and handshake stability.
- **Application Intent (Layer 7)**: Analyzing protocol-specific payloads for malicious activity.
- **Lateral Movement**: Tracking how threats pivot between internal stack layers.

The Light Touch Approach

- **Header Priority**: Focusing on Flags and Sequence Numbers to map the session.
- **Cognitive Optimization**: Analyzing protocol-specific payloads for malicious activity.
- **Protocol Scoping**: Isolating traffic by port and service before deep packet inspection.

Technical Mapping

- **Encapsulation Check**: Verifying that data segments match expected protocol.
- **Anomaly Detection**: Identifying non-standard behavior within specific stack levels.
- **Isolation Strategy**: Narrowing the scope of a hunt to a single layer of the stack.

Why We Filter

Transforming the Firehose into Evidence

The Noise vs Signal Problem

- **Protocol Elimination**: Suppressing high-volume background noise like ARP, DHCP, and SSDP.
- **Service Filtering**: Masking local discovery traffic (LLMNR/NBNS) to reveal external intent.
- **Resource Management**: Optimizing hardware performance by reducing the active packet set.

Operational Benefits

- **Artifact Velocity**: Drastically reducing the time to locate specific malicious indicators.
- **Narrative Clarity**: Creating clean views that simplify complex attack chains.
- **Evidence Preservation**: Isolating relevant packets for streamlined reporting and documentation.

Precision Analysis

- **Dynamic Scoping**: Adjusting filters in real-time as the investigation pivots.
- **False Positive Reduction**: Hardening the dataset against common network chatter.
- **Logic-Based Hunting**: Using Boolean operators to isolate specific session attributes.

TCP Filtering | Analyzing Handshakes and Session State

Connection Integrity

- **Handshake Validation**: Verifying successful three-way handshakes (SYN, SYN-ACK, ACK).
- **Retransmission Analysis**: Identifying packet loss, latency, or potential interference.
- **State Tracking**: Monitoring the lifecycle of a session from establishment to teardown.
- **Flow Control**: Observing window size scaling and congestion notification.

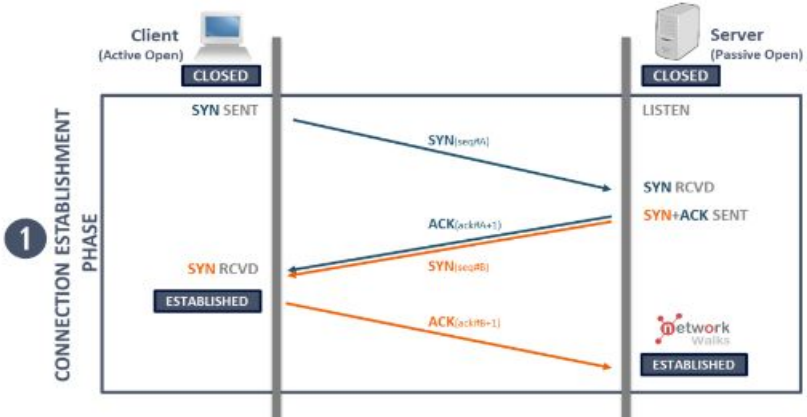
Flag-Based Syntax

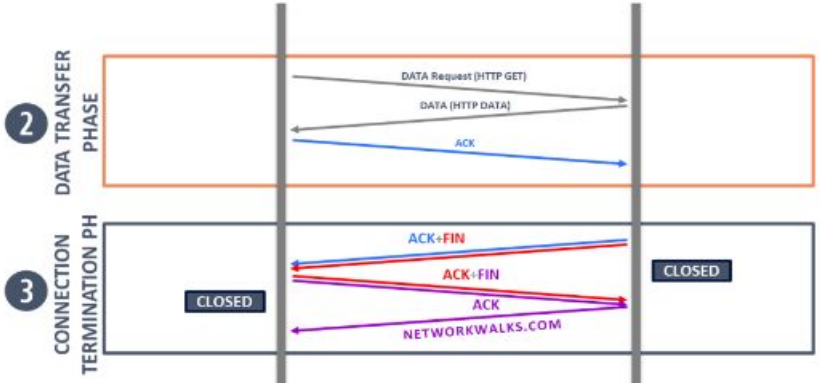
- `tcp.flags.syn == 1`: Isolates initial connection attempts.
- `tcp.flags.reset == 1`: Identifies abruptly terminated or refused connections.
- `tcp.flags.fin == 1`: Highlights graceful session closures.
- `tcp.flags.push == 1`: Locates immediate data transfer bursts within a stream.

Performance & Error Logic

- `tcp.analysis.retransmission`: Pinpoints network congestion or dropped packets.
- `tcp.analysis.zero_window`: Detects receiver-side processing bottlenecks.
- `Tcp.analysis.lost_segment`: Maps gaps in the sequence number stream.

TCP 3-WAY HANDSHAKE PROCESS





UDP & DNS

Investigating Connectionless Traffic and Name Resolution

UDP Syntax & Identification

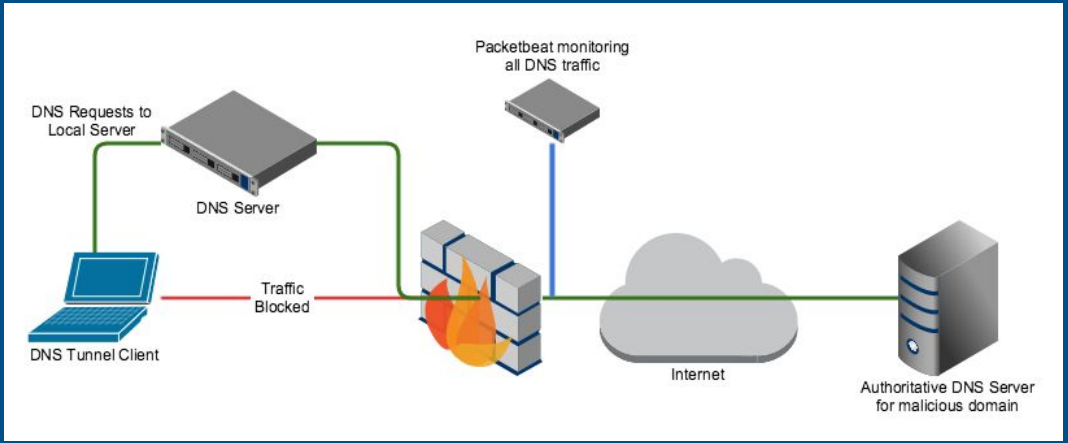
- **Non-Standard Traffic**: `udp.port != 53` and `udp.port != 123` to isolate atypical UDP behavior.
- **String Complexity**: `dns.qry.name.len > 20` to catch unusually long, suspicious domains.
- **Tunneling Detection**: Identifying `dns.count.labels > 4` for deeply nested, tunneled traffic.
- **Payload Inspection**: Filtering for `dns.txt > 100` to find large, script-heavy records.

DNS Forensic Artifacts

- **Outbound Queries**: Tracking `dns.flags.response == 0` to audit internal host requests.
- **High-Entropy Subdomains**: Identifying randomized strings that signal encoded data transfer.
- **C2 Communication**: Spotting TXT records used as vessels for remote commands.
- **Response Analysis**: Monitoring NXDOMAIN spikes indicative of subdomain brute-forcing.

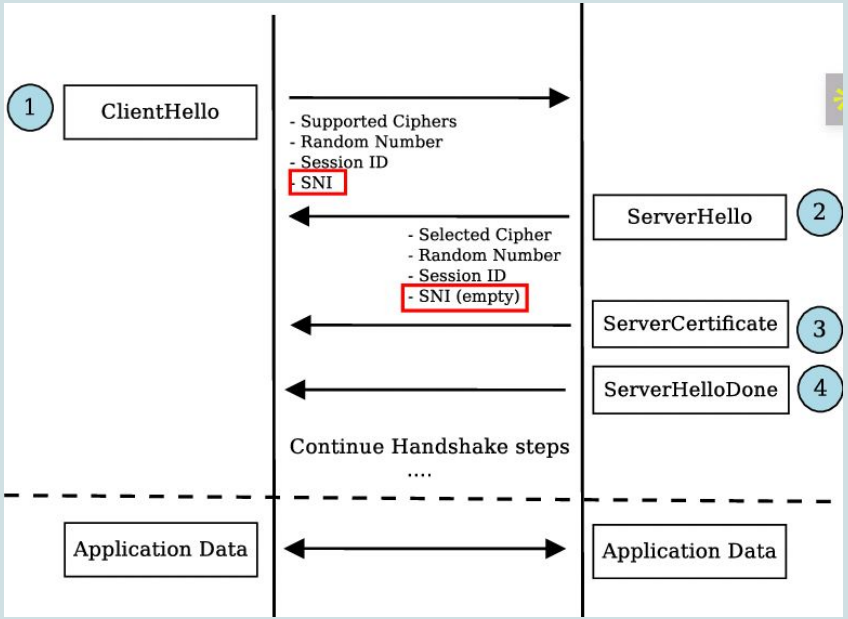
Operational Context

- **Stateless Risk**: Managing the lack of handshake verification in connectionless protocols.
- **Vessel Activity**: Recognizing DNS as a high-trust protocol often ignored by firewalls.
- **Data Exfiltration**: Visualizing how small, fragmented packets bypass volume-based alerts.



HTTP/S Analysis

Analyzing Web Traffic and Encrypted Indicators



Application Layer Indicators

- **Data Submission**: `http.request.method == "POST"` to audit outbound data transfers.
- **User-Agent Analysis**: Identifying automated tools like curl, PowerShell, or Python requests.
- **Path Investigation**: Monitoring for suspicious URI strings and unauthorized file access.
- **Response Codes**: Tracking 403 Forbidden or 404 Not Found for directory brute-forcing.

Encrypted Traffic Visibility

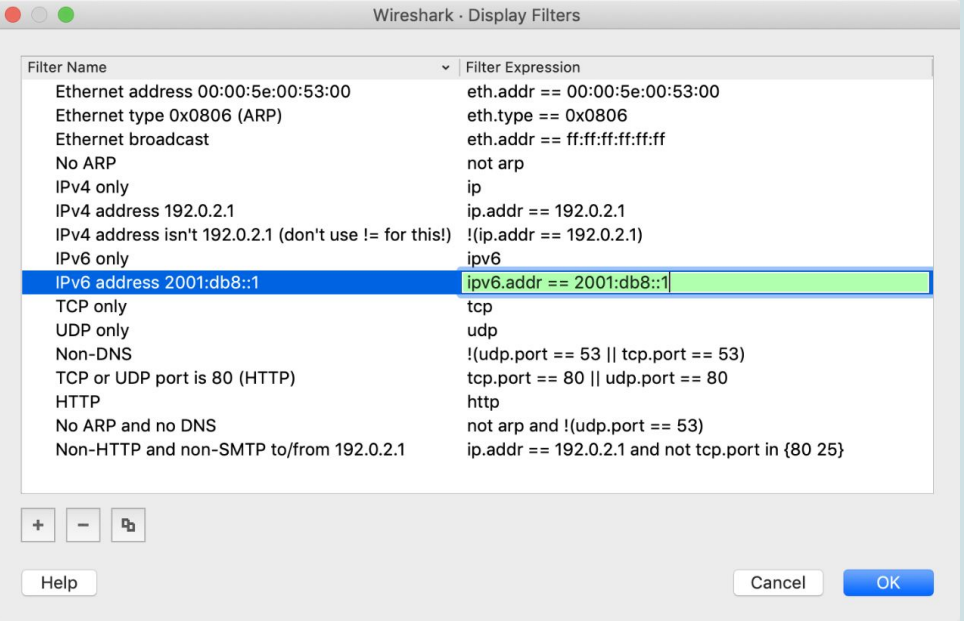
- **Domain Identification**: `tls.handshake.extensions_server_name` to extract the target via SNI.
- **Certificate Validation**: Detecting self-signed certificates or anomalous issuer names.
- **Protocol Hygiene**: Identifying legacy TLS 1.0/1.1 versions and weak cipher suites.
- **Fingerprinting**: Utilizing JA3 hashes to identify specific client-side malware signatures.

Secure Channel Analysis

- **Handshake Integrity**: Verifying the Client Hello and Server Hello exchange.
- **Certificate Pinning**: Spotting discrepancies in expected vs presented public keys.
- **Traffic Entropy**: Distinguishing between standard encrypted web traffic and tunneled C2.

Combining Filters

Surgical Navigation via Boolean Logic



Logical Operators

- **AND Usage**: Narrowing results to specific, interesting criteria.
- **OR Usage**: Expanding results to include multiple distinct possibilities.
- **NOT Usage**: Excluding known-safe traffic (such as `not arp` or `not ssl`.
- **Nesting**: Using parentheses to prioritize complex logical evaluations.

Precision Example

- **Target Isolation**: `ip.addr == <Target> and not tcp.port == 443`
- **Exclusion Logic**: Removing noise to reveal non-standard protocol usage.
- **Refined Scope**: Combining IP, port, and flag filters for surgical accuracy.

Operation Workflow

- **Noise Reduction**: Progressively stripping away authorized traffic layers.
- **Pivot Discovery**: Using inclusive logic to follow an attacker across paths.
- **Result Validation**: Testing filter strings to ensure critical data is not eclipsed.

The Big Picture

Visualizing Host-to-Host Relationships

Address A	Address B	Packets	Bytes	Total Packets	Percent filtered	Packets A → B	Bytes A → B	Packets B → A	Packets B → A	Rel Start	Du
0.0.0.0	255.255.255.255	1	383 bytes	1	100.00%	1	383 bytes	0	0 bytes	34.656060	0
142.250.186.67	192.168.1.85	4	301 bytes	30	13.33%	2	148 bytes	2	153 bytes	31.378530	1
142.250.186.165	192.168.1.85	28	12,467 KiB	29	96.55%	16	8,460 KiB	12	4,007 KiB	0.000000	0
152.199.19.160	192.168.1.85	7	543 bytes	7	100.00%	4	345 bytes	3	198 bytes	7.470308	0
162.159.134.234	192.168.1.85	43	5,452 KiB	43	100.00%	22	4,239 KiB	21	1,213 KiB	8.030960	87
172.66.251.78	192.168.1.85	204	23,745 KiB	204	100.00%	102	15,616 KiB	102	8,129 KiB	1.374770	102
172.104.245.130	192.168.1.85	66	5,221 KiB	66	100.00%	35	2,559 KiB	31	2,662 KiB	5.196435	90
172.217.18.106	192.168.1.85	29	3,502 KiB	29	100.00%	16	1,716 KiB	13	1,786 KiB	12.638637	81
192.168.1.1	192.168.1.255	10	600 bytes	10	100.00%	10	600 bytes	0	0 bytes	6.083682	90
192.168.1.1	224.0.0.251	329	56,480 KiB	329	100.00%	329	56,480 KiB	0	0 bytes	0.555528	103
192.168.1.1	255.255.255.255	12	720 bytes	12	100.00%	12	720 bytes	0	0 bytes	0.862892	102
192.168.1.21	192.168.1.255	10	1,295 KiB	10	100.00%	10	1,295 KiB	0	0 bytes	1.476232	97.
192.168.1.21	255.255.255.255	4	792 bytes	4	100.00%	4	792 bytes	0	0 bytes	8.848800	90
192.168.1.35	224.0.0.251	1	279 bytes	1	100.00%	1	279 bytes	0	0 bytes	43.258929	0
192.168.1.35	239.255.255.250	93	50,372 KiB	93	100.00%	93	50,372 KiB	0	0 bytes	19.908648	82
192.168.1.36	224.0.0.251	2	1,633 KiB	2	100.00%	2	1,633 KiB	0	0 bytes	39.577346	25
192.168.1.54	192.168.1.85	2	120 bytes	2	100.00%	1	54 bytes	1	66 bytes	33.074264	0.
192.168.1.54	224.0.0.251	1	326 bytes	1	100.00%	1	326 bytes	0	0 bytes	29.124620	0.
192.168.1.85	17.248.145.106	55	28,001 KiB	55	100.00%	26	11,066 KiB	29	16,936 KiB	53.787326	19
192.168.1.85	31.13.84.51	17	1,813 KiB	17	100.00%	10	1,011 KiB	7	822 bytes	3.847714	86
192.168.1.85	31.13.84.52	21	1,750 KiB	21	100.00%	14	1,107 KiB	7	658 bytes	2.200680	60
192.168.1.85	35.186.224.25	19	2,550 KiB	43	44.19%	8	782 bytes	11	1,786 KiB	13.462394	90
192.168.1.85	35.186.224.40	16	1,355 KiB	16	100.00%	8	700 bytes	8	688 bytes	10.725210	90
192.168.1.85	35.186.224.47	29	5,505 KiB	29	100.00%	14	1,143 KiB	15	4,362 KiB	17.476152	79
192.168.1.85	54.216.252.255	15	1,230 KiB	15	100.00%	10	805 bytes	5	455 bytes	3.428718	100
192.168.1.85	104.199.65.124	5	363 bytes	5	100.00%	3	209 bytes	2	154 bytes	40.114078	59
192.168.1.85	136.143.190.75	40	9,621 KiB	40	100.00%	23	2,975 KiB	17	6,646 KiB	0.855213	100
192.168.1.85	142.250.186.74	83	39,547 KiB	83	100.00%	41	10,481 KiB	42	29,066 KiB	53.776691	1
192.168.1.85	172.217.18.1	86	49,088 KiB	86	100.00%	42	10,907 KiB	44	38,181 KiB	54.327884	30.
192.168.1.85	192.168.1.21	9	670 bytes	9	100.00%	5	390 bytes	4	280 bytes	1.477461	97.
192.168.1.85	192.168.1.32	1	46 bytes	1	100.00%	1	46 bytes	0	0 bytes	20.597148	0.
192.168.1.85	192.168.1.250	36	3,768 KiB	36	100.00%	18	1,433 KiB	18	2,335 KiB	14.918659	39

The Conversations Window

- **Aggregation Strategy**: Navigating to `Statistics > Conversation` for sesion-level data.
- **Logical Grouping**: Consolidating thousands of individual packets into distinct, readable flows.
- **Protocol Scoping**: Sorting by Ethernet, IP, TCP, or UDP to isolate the attack vector.
- **Efficient Mapping**: Visualizing the entire network landscape from a single management view.

Top-Level Metrics

- **Host Identification**: Determining the primary talkers (Internal vs External endpoints).
- **Data Volume**: Evaluating Byte Count to identify potential exfiltration or heavy downloads.
- **Temporal Analysis**: Analyzing session duration to distinguish brief scans from persistent links.
- **Packet Symmetry**: Comparing sent vs received counts to identify Push or Pull behavior.

Forensic Significance

- **Outlier Detection**: Spotting high-bandwidth sessions that deviate from established baselines.
- **Geographic Triage**: Checking external IP ownership for unauthorized or high-risk regions.
- **Infection Tracking**: Identifying Patient-Zero by tracing the earliest timestamped flows.

Endpoint Evaluation

Identifying Top Talkers and Asset Risks

Internal vs External Mapping

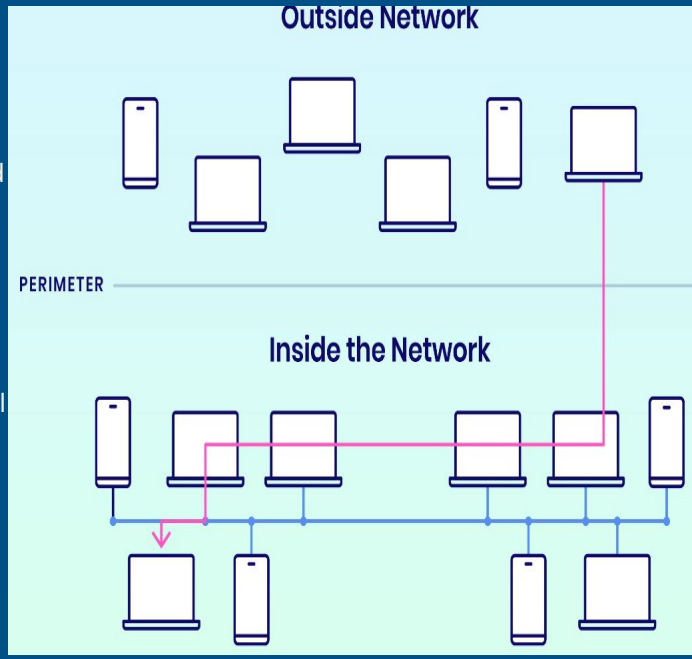
- **Infrastructure Attribution**: Tracking connections to known VPS providers, such as DigitalOcean, AWS, or Linode.
- **Service Exposure**: Auditing internal hosts reaching out via Port 445 (SMB) or Port 3389 (RDP).
- **Geographic Triage**: Identifying traffic bound for high-risk or unexpected international IP ranges.
- **Beacon Identification**: Spotting persistent, low-volume connections to external command nodes.

Lateral Movement Artifacts

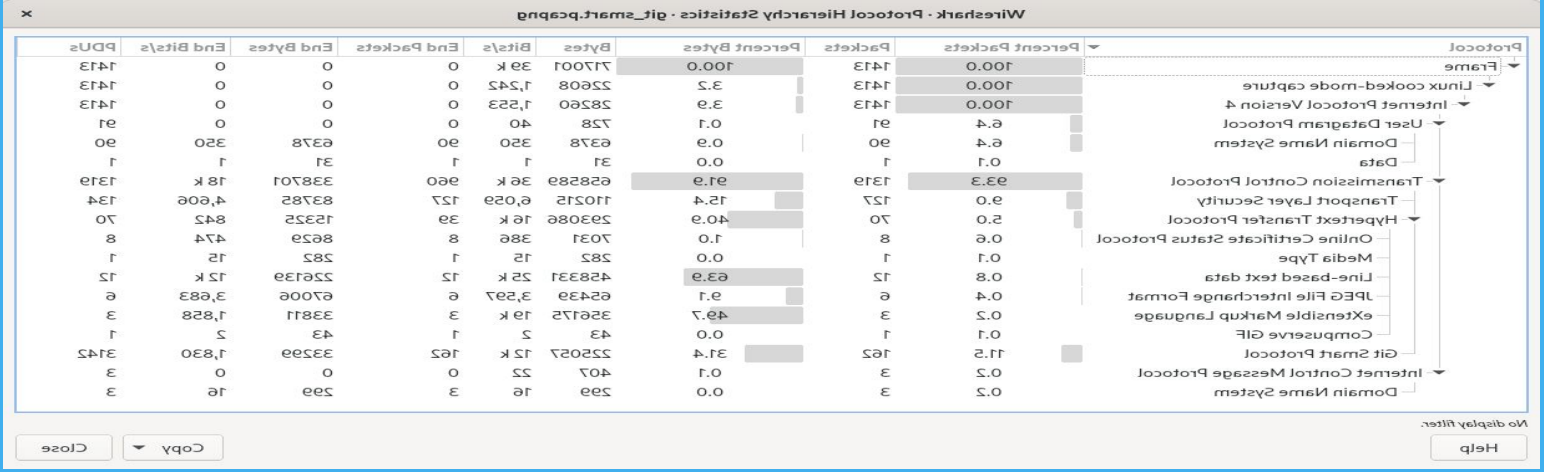
- **Administrative Spikes**: Monitoring high volumes of internal-to-internal SMB or RPC traffic.
- **Peer-to-Peer Anomalies**: Flagging workstations communicating directly with other workstations.
- **Credential Hopping**: Identifying repeated authentication attempts across multiple internal assets.
- **Protocol Misuse**: Detecting non-standard administrative tools moving through the local network.

Asset Risk Profiling

- **Criticality Assessment**: Prioritizing investigations based on the sensitivity of the involved internal hosts.
- **Baselines**: Comparing current endpoint behavior against historical normal traffic patterns.
- **Blast Radius**: Visualizing how many internal systems a compromised host has touched.



Protocol Hierarchy | Validating Traffic Distribution vs Network Baseline



Protocol Mismatch Detection

- **Anomalous Ports**: Spotting cleartext Telnet or HTTP behavior masquerading over Port 443.
- **Encapsulation Errors**: Identifying non-DNS traffic attempting to egress via Port 53.
- **Service Validation**: Verifying that the protocol structure matches the assigned port designation.
- **Tunnels and Wrappers**: Detecting SSH or VPN signatures inside standard web traffic streams.

Visualizing the Mix

- **Hierarchical Mapping**: Navigating to `Statistics > Protocol Hierarchy` for a top-down view of the capture.
- **Percentage Distribution**: Evaluating the ratio of TCP vs UDP to establish a behavioral baseline.
- **Unauthorized Discovery**: Identifying non-business protocols like IRC, BitTorrent, or Tor.
- **Sub-Protocol Auditing**: Drilling down into specific application layers to find hidden payloads.

Defining Normal

The Necessity of the Baseline

Network Identity

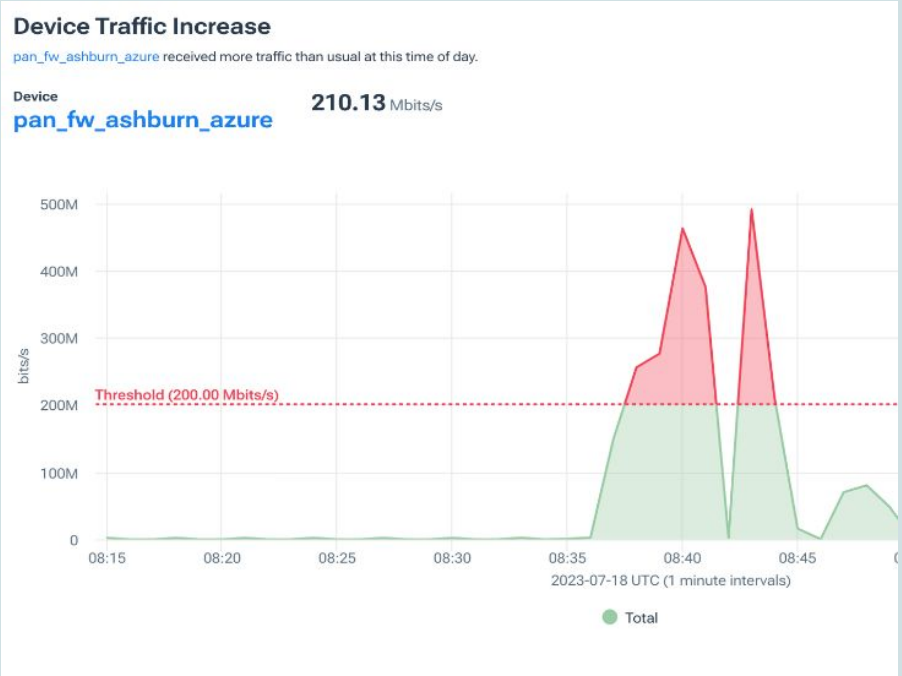
- **Temporal Patterns**: Documenting expected business-hour spikes to separate commerce from compromise.
- **Authorized Assets**: Mapping known Service Accounts to administrative IP ranges.
- **Protocol Whitelisting**: Establishing a Golden Image of standard application traffic.
- **User Behavior**: Identifying the typical volume and destination of standard department workflows.

Anomalous Timing

- **Temporal Shifting**: Identifying after-hours data bursts that suggest automated exfiltration.
- **Unauthorized Flows**: Spotting new, internal-to-internal connections during low-staffing periods.
- **Execution Windows**: Correlating network spikes with scheduled tasks vs manual attacker activity.

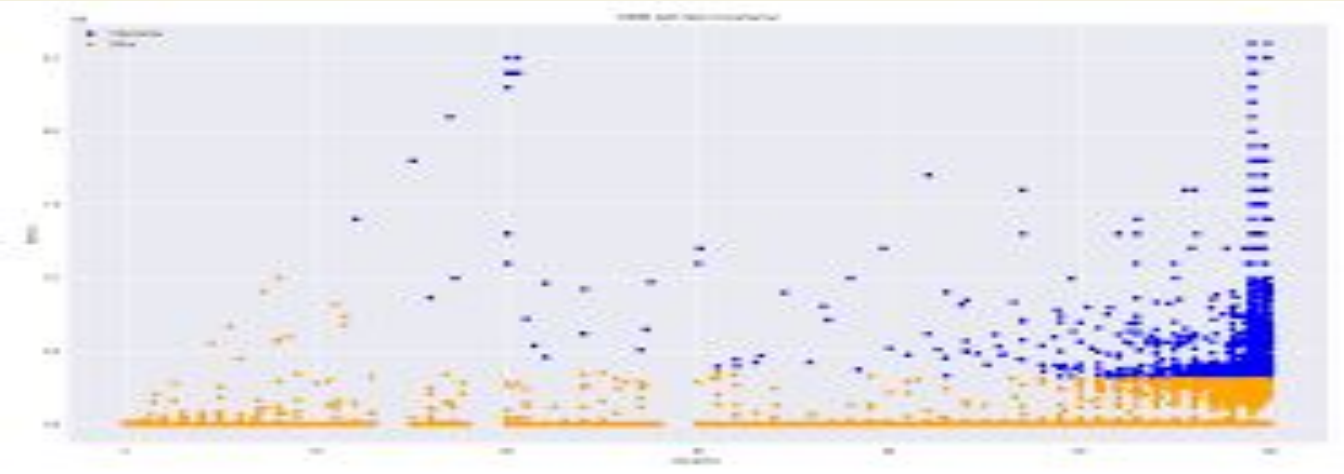
Deviation Analysis

- **Volume Thresholds**: Setting alerts for byte counts that exceed the historical daily average.
- **New Endpoint Discovery**: Flagging the first appearance of an external IP within the environment.
- **Ratio Shifts**: Monitoring for sudden changes in the Upload-to-Download balance.



Elephant vs Mice Flows

Volume Analysis for Threat Detection



Elephant Flows (Volume-Based)

- **Massive Transfers**: Identifying sustains, high-bandwidth outbound flows on FTP, SFTP, or HTTPS.
- **Exfiltration Risk**: Spotting large-scale data staging or theft in progress.
- **Network Impact**: Detecting saturation that degrades service for legitimate traffic.
- **Symmetry Check**: Monitoring for high Push ratios with minimal inbound responses.

Mice Flows (Frequency-Based)

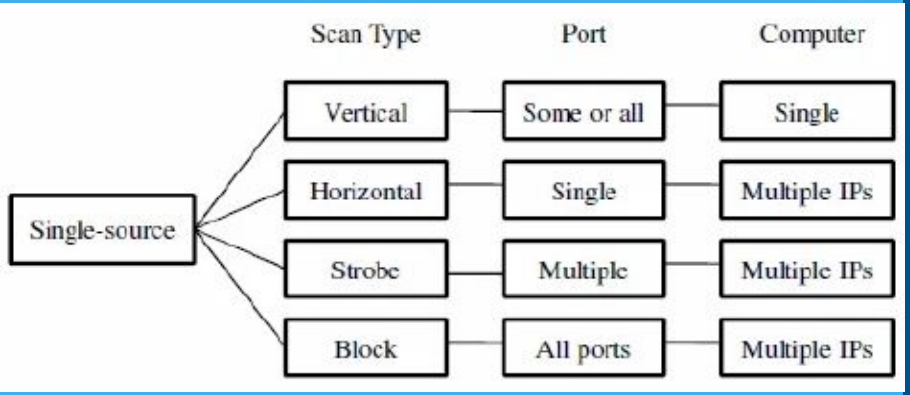
- **C2 Heartbeats**: Detecting sustained, low-volume packets with minimal or empty payloads.
- **Keep-Alives**: Identifying persistent connections designed to bypass firewall timeouts.
- **Low-and-Slow**: Recognizing automated queries that evade volume-based detection thresholds.
- **Consistency**: Analyzing Time-Series patterns for lack of human-generated jitter.

Analytical Differentiation

- **Volume vs Velocity**: Distinguishing between a single large Heist and consistent Whispers.
- **Triage Strategy**: Prioritizing investigations based on the immediate impact to data integrity.
- **Signature Mapping**: Matching flow characteristics to known malware communication profiles.

Port Scanning Patterns

Recognizing Reconnaissance Signatures



Scan Types

- **Vertical Scan**: A single source IP probing multiple ports on a specific host to find vulnerabilities.
- **Horizontal Sweep**: A single source IP targeting one specific port across an entire subnet.
- **Strobe Scan**: Focusing on a small set of known-vulnerable ports across multiple systems.
- **Target Mapping**: Identifying the inventory phase before an exploit is launched.

Forensic Artifacts

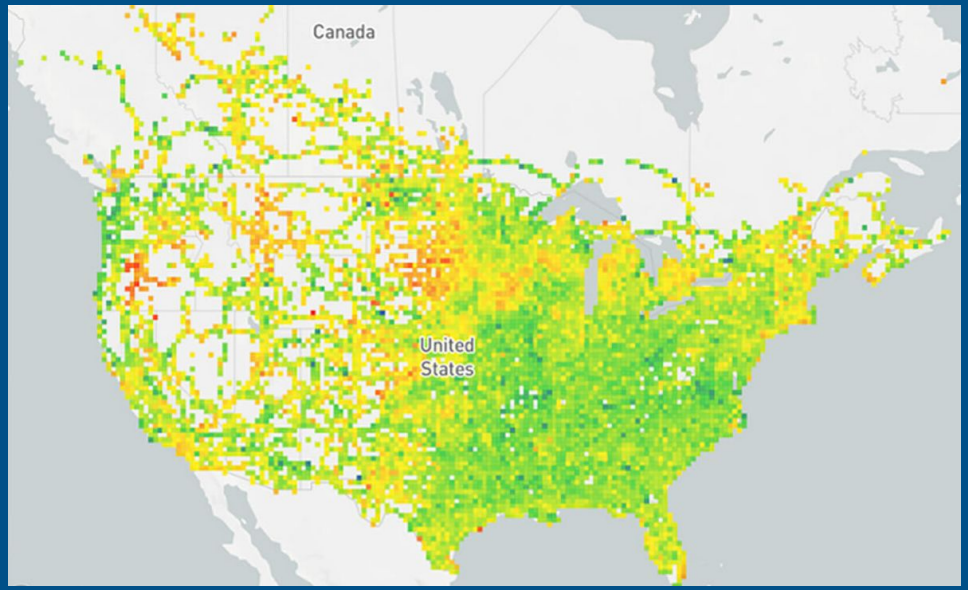
- **Port Unreachable**: Monitoring for ICMP Type 3, Code 3 responses from the target host.
- **SYN-Steal Signatures**: Identifying half-open connections that never complete the handshake.
- **Reset Spikes**: Tracking high volumes of RST packets in response to closed-port probes.
- **Sequence Timing**: Detecting rapid-fire query intervals that lack huma-like delay.

Defensive Context

- **Triage Priority**: Distinguishing between internet background noise and targeted internal scans.
- **Honeypot Hits**: Identifying probes against non-existent or high-value internal decoy assets.
- **Scanning Direction**: Determining if the scan is originating from an external actor or a compromised internal host.

Geolocation Anomalies

Mapping IP Addresses to Physical Locations



Contextual Analysis

- **Business Presence**: Flagging traffic to regions where the organization has no employees or customers.
- **Database Integration**: Utiziling MaxMind or IPinfo for real-time geographic insights.
- **Policy Alignment**: Correlating traffic destinations with corporate Geofencing restrictions.
- **Travel Profiling**: Comparing user login locations against known employee travel itineraries.

Risk Assessment

- **Host Reputation**: Investigating Traffic to jurisdictions known for Bulletproof hosting.
- **VPN/Proxy Detection**: Identifying exit nodes used to mask an attacker's true origin.
- **Data Sovereignty**: Monitoring for unauthorized data movement across international borders.
- **High-Risk Zones**: Prioritizing alerts for connections originating from sanctioned or high-threat regions.

Forensic Significance

- **Impossible Travel**: Detecting logins from two distant locations within a physically impossible timeframe.
- **Inbound Spikes**: Tracking sudden increases in unsolicited traffic from a specific country.
- **Exfiltration Paths**: Mapping the final destination of stolen data packets for attribution.

What is a Beacon? | Malware Command & Control (C2) Persistence

The C2 Heartbeat

- **Automated Check-ins**: Periodic connections to a remote server to retrieve new instructions.
- **Protocol Blending**: Disguising malicious traffic within standard HTTPS or DNS streams.
- **Instruction Retrieval**: Pulling down shell commands, scripts, or additional payloads.
- **Persistence**: Maintaining a foothold even after system reboots or network changes.

User vs Machine

- **Mechanical Regularity**: Identifying exact, repeating intervals (every five minutes, for example) that signal a script.
- **Human Jitter**: Distinguishing sporadic browsing habits from automated software patterns.
- **Temporal Entropy**: Measuring the randomness of connection times to find hidden automation.
- **Payload Consistency**: Spotting identical request sizes that suggest a programmed Keep-Alive.

Detection Markers

- **Long-Tail Analysis**: Reviewing 24-hour captures to find connections that never stop.
- **Domain Reputation**: Checking if the beaconing endpoint is a newly registered domain (NRD).
- **Header Staticity**: Finding identical User-Agents and Cookie strings across every request.

Visualizing Timing

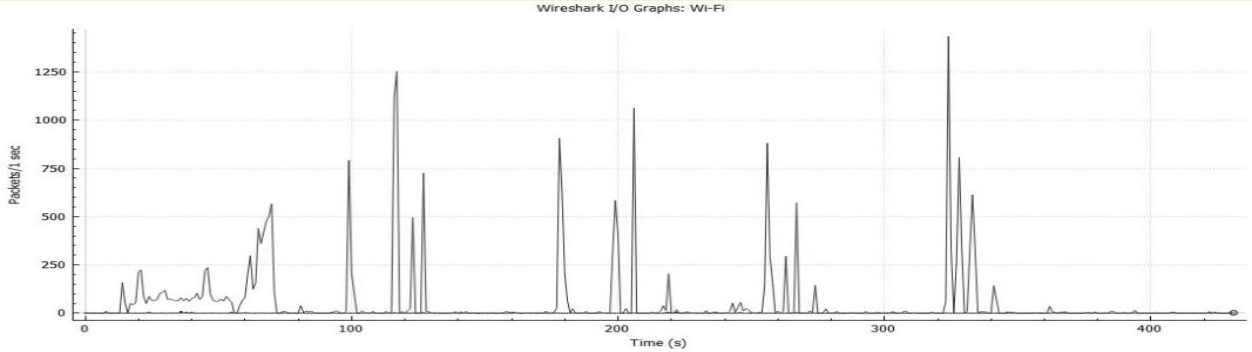
Time-Delta Analysis

Syntax & Setup

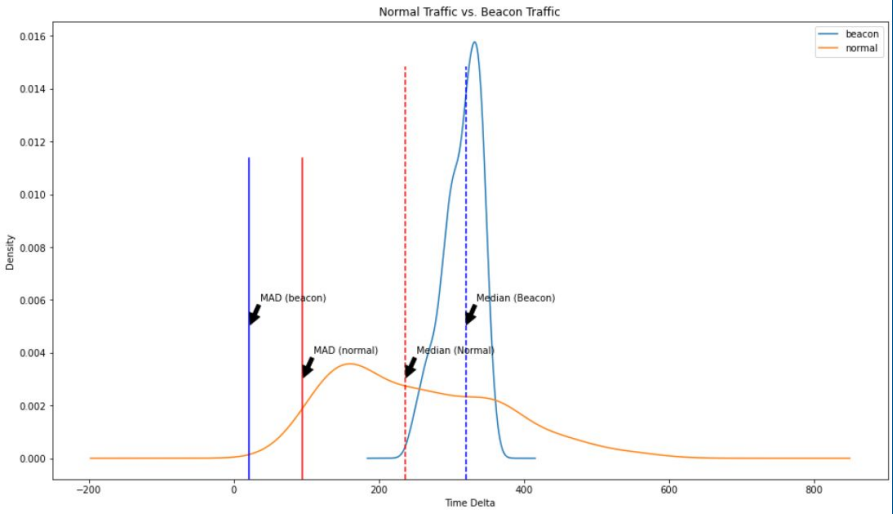
- **Column Configuration**: Adding `frame.time_delta_displayed` to the packet list for precision.
- **Perfect Periodicity**: Finding packets spaced by exactly 60.000s or other fixed intervals.
- **Precision Tracking**: Measuring the microsecond differences between related request packets.
- **Filtering Logic**: Isolating specific streams to calculate the gap between C2 check-ins.

Graphical Analysis

- **IO Graphs**: Navigation to `Statistics > IO Graphs` to map packet frequency over time.
- **Interval Clumping**: Visualizing bursts of traffic occurring at predictable, fixed cycles.
- **Jitter Evaluation**: Identifying intentional delays added by malware to mimic human noise.
- **Baseline**: Comparing suspicious timing against known, legitimate system updates.



The Jitter Factor | Evading Simple Detection via Randomness



Timing Variance

- **Jitter Definition**: Randomly adding or subtracting time to break a perfectly fixed interval.
- **The ±20% Window**: Hiding the C2 heartbeat within a variable time range to mimic human unpredictability.
- **Evasion Strategy**: Defeating simple automated alerts that only flag exact periodicity (such as 60.000s).
- **Pattern Masking**: Blending malicious check-ins with the organic noise of standard web browsing.

Mathematical Detection

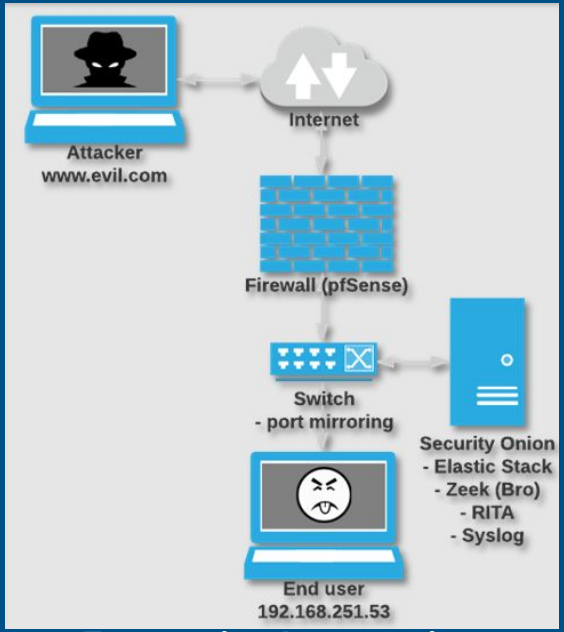
- **Average Intervals**: Calculating the mean time-delta to identify a baseline frequency despite the jitter.
- **Cluster Analysis**: Grouping packets by destination to see if clumps still occur within a specific window.
- **Long-Tail Observation**: Monitoring traffic over 24-48 hour to reveal the persistent underlying script.
- **Standard Deviation**: Using math to prove that the randomness is still too consistent for a human user.

Tools for Beacon Detection

Automation Beyond Manual Analysis

Statistical Tools

- **RITA (Real Intelligence Threat Analytics)**: Open-source framework for detecting C2 through beaconing and long-duration connections.
- **Interval Tagging**: Utilizing specialized Wireshark Lua scripts to automate the calculation of time-deltas across large pcaps.
- **Score-Based Analysis**: Applying mathematical models to rank connections based on their likelihood of being automated.
- **Open Source Ecosystem**: Leveraging community-driven tools to augment manual packet inspection.

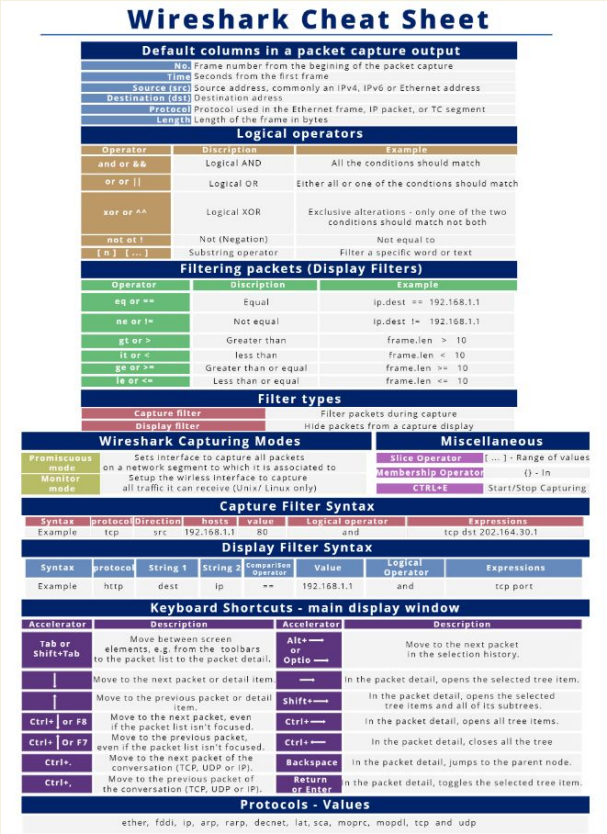


Enterprise Integration

- **Zeek (Formerly Bro)**: Utilizing high-level network logs for long-term pattern identification without the overhead of full packet captures.
- **SIEM Correlation**: Exporting metadata to platforms like Splunk or ELK for cross-protocol behavioral analysis.
- **Fleet Visibility**: Monitoring thousands of endpoints simultaneously to find top-talkers in real-time.
- **Data Retention**: Analyzing trends over weeks or months to catch low-and-slow actors that evade short-term windows.

The Analyst's Mindset

Objectivity and Evidence-Based Logic



Wireshark Cheat Sheet

Default columns in a packet capture output

No.	Frame number from the begining of the packet capture
Time	Seconds from the first frame
Source (src)	Source address, commonly an IPv4, IPv6 or Ethernet address
Destination (dst)	Destination adress
Protocol	Protocol used in the Ethernet frame, IP packet, or TC segment
Length	Length of the frame in bytes

Logical operators

Operator	Discription	Example
and or &&	Logical AND	All the conditions should match
or or ||	Logical OR	Either all or one of the condtions should match
xor or ^^	Logical XOR	Exclusive alterations - only one of the two conditions should match not both
not or !	Not (Negation)	Not equal to
[n] [...]	Substring operator	Filter a specific word or text

Filtering packets (Display Filters)

Operator	Discription	Example
eq or ==	Equal	ip.dest == 192.168.1.1
ne or !=	Not equal	ip.dest != 192.168.1.1
gt or >	Greater than	frame.len > 10
lt or <	less than	frame.len < 10
ge or >=	Greater than or equal	frame.len >= 10
le or <=	Less than or equal	frame.len <= 10

Filter types

Capture filter	Filter packets during capture
Display filter	Hide packets from a capture display

Wireshark Capturing Modes

Promiscuous mode	Sets interface to capture all packets on a network segment to which it is associated to
Monitor mode	Setup the wirless interface to capture all traffic it can receive (Unix/ Linux only)

Miscellaneous

Slice Operator	[...] - Range of values
Membership Operator	{} - In
CTRL+E	Start/Stop Capturing

Capture Filter Syntax

Syntax	protocol	Direction	hosts	value	Logical operator	Expressions
Example	tcp	src	192.168.1.1	80	and	tcp dst 202.164.30.1

Display Filter Syntax

Syntax	protocol	String 1	String 2	Comparison Operator	Value	Logical Operator	Expressions
Example	http	dest	ip	==	192.168.1.1	and	tcp port

Keyboard Shortcuts - main display window

Accelerator	Description	Accelerator	Description
Tab or Shift+Tab	Move between screen elements, e.g. from the toolbars to the packet list to the packet detail.	Alt+→ or Optio→	Move to the next packet in the selection history.
↓	Move to the next packet or detail item.	→	In the packet detail, opens the selected tree item.
↑	Move to the previous packet or detail item.	Shift+→	In the packet detail, opens the selected tree items and all of its subtrees.
Ctrl+↓ or F8	Move to the next packet, even if the packet list isn't focused.	Ctrl+→	In the packet detail, opens all tree items.
Ctrl+↑ Or F7	Move to the previous packet, even if the packet list isn't focused.	Ctrl+←	In the packet detail, closes all the tree
Ctrl+.	Move to the next packet of the conversation (TCP, UDP or IP).	Backspace	In the packet detail, jumps to the parent node.
Ctrl+,	Move to the previous packet of the conversation (TCP, UDP or IP).	Return or Enter	In the packet detail, toggles the selected tree item.

Protocols - Values

ether, fddi, ip, arp, rarp, decnet, lat, sca, moprc, mopdl, tcp and udp

Discipline over Hunch

- **Data-Driven Narratives**: Letting the packets tell the story rather than forcing a preconceived theory.
- **Bias Mitigation**: Actively avoiding Confirmation Bias by seeking evidence that disproves your own hypothesis.
- **The Zero Trust Lens**: Treating every internal host as a potential threat until the traffic proves otherwise.
- **Rigorous Validation**: Cross-referencing findings across multiple tools (such as Wireshark and Zeek logs).

The Investigation Trail

- **Analyst Notebooks**: Maintaining a chronological log of applied filters, timestamps, and specific packet numbers.
- **Reproducibility**: Ensuring that another analyst can apply your filter strings and reach the same conclusion.
- **Documentation Hygiene**: Capturing screenshots and exporting specific packet ranges for evidence preservation.
- **Knowledge Transfer**: Building a library of successful hunt queries to share with the wider SOC team.

Documentation & Mapping | Aligning Evidence to Defensive Frameworks

MITRE ATT&CK Correlation

- **T1071 (Application Layer Protocol)**: Identifying malware using standard protocols (HTTP/DNS) to blend in with web traffic.
- **T1041 (Exfiltration Over C2 Channel)**: Mapping high-volume outbound Elephant flows to stolen data transfers.
- **T1046 (Network Service Scanning)**: Cataloging reconnaissance signatures like horizontal sweeps or vertical port scans.
- **T1571 (Non-Standard Port)**: Documenting protocols operating outside of their assigned port designations.

Reporting Artifacts

- **File Integrity**: Including the SHA256 hash of the original PCAP to ensure evidence remains untampered.
- **Precision Referencing**: Documenting specific Frame Numbers for every critical packet cited in the report.
- **Evidence Export**: Attaching exported `.pcapng` snippets containing only the relevant malicious streams.
- **File Transparency**: Listing the exactly Wireshark display filters used to reach the reported conclusions.

Defensive Posture

- **Control Validation**: Using packet data to prove if existing firewalls or EDRs successfully blocked an action.
- **Gap Analysis**: Identifying Blind Spots where logging or inspection failed to catch a specific technique.
- **Continuous Improvement**: Feeding captured signatures back into the SIEM for future automated alerting.

Summary & Resources | The 5-Step Process Review

The Pillars

- **Filter**: Systematically reducing millions of packets into a manageable, targeted dataset.
- **Analyze**: Evaluating protocol behavior, timing, and session integrity for anomalies.
- **Identify**: Pinpointing Top-Talkers, internal risks, and compromised assets.
- **Recognize**: Mapping patterns like Beaconing, Port Scanning, and Data Exfiltration.
- **Document**: Translating technical findings into evidence-based, framework-aligned reports.

The Analyst's Path

- **Continuous Practice**: Regularly analyzing Normal traffic to sharpen your eye for the Abnormal.
- **Community Engagement**: Share findings on a platform of choice (LinkedIn, X, Skool, etc.) to refine your defensive logic and receive feedback.
- **Tool Fluency**: Moving beyond basic filters into advanced statistics, IO graphing, and Lua scripting.

Expert Video & Documentation

- **Christ Greer (Packet Analysis Playlist)**: https://www.youtube.com/playlist?list=PLW8bTPfXNGdA_TprronpuNh7Ei8imYppX
- **Wireshark User Guide**: https://www.wireshark.org/docs/wsug_html_chunked/

Learning Resources

- **TryHackMe**: https://tryhackme.com/hacktivities/search?page=1&kind=all&searchText=wireshark&order=relevance
- **Labex**: https://labex.io/learn/wireshark
- **LetsDefend**: https://app.letsdefend.io/training/lesson_detail/installing-the-wireshark
- **BlueTeamLabs.online (Piggy)**: https://blueteamlabs.online/home/investigation/piggy-aij2bd8h2

Practical Quiz:Operation: SANDSTONES

A Deep-Dive Analysis of the TrickBot “rob13” Infection.

Phase One: Initial Compromise & Host ID

1. When attempting to identify the first stage of an infection (the external downloader), which query would you use?

A. `http.request`
B. `http.uri == “rob13”`
C. `http_request_method == GET`
D. `ip.addr contains “malicious”`

2. After filtering for the initial request, what is the Source IP of the infected victim machine?

A. 10.2.17.2
B. 192.168.1.1
C. 10.2.17.101
D. 172.16.0.50

3. Looking at the first HTTP request, what is the malicious Domain (URL) the victim is communicating with?

A. checkip[.]amazonaws[.]com
B. Destinostumundo[.]com
C. Sandstones[.]local
D. microsoft-update[.]com

Phase One: Initial Compromise & Host ID Continued

4. In that same HTTP request, what is the specific URI (path) being requested from the server?

A. .rob13/90/gate.php
B. /index.html
C. /admin/login.php
D. /layout/recruiter.php

5. To identify the NetBIOS name of the infected host (10.2.17.101), which query is the most efficient?

A. `ip.src == 10.2.17.101 && nbns`
B. `ip.addr == 10.2.17.101 && dhcp`
C. `nbns.name_contains == 10.2.17.101`
D. `find.host == “10.2.17.101”`

5. After running the NBNS query, what is the confirmed hostname of the infected machine?

A. DESKTOP-5BT9E19
B. DESKTOP-ADIJBT3
C. SANDSTONES-DC
D. WIN-SERV-2019

Phase Two: DNS & Infrastructure Discovery

7. In order to find how the host located the Domain Controller (DC) via Advice Directory Service Records, which query should we run?

A. `dns.flags.response == 1`
B. `dns_type == SRV`
C. `dns.qry.type == 33`
D. `ip.proto == DNS & srv.record`

8. Looking at the SRV records, which internal IP address is identified as the Domain Controller for the SANDSTONES domain?

A. 10.2.17.1
B. 10.2.17.101
C. 10.2.17.255
D. 10.2.17.2

Phase 3: C2 Triage & Behavior

9. When triaging top-talkers in the conversations window, why is the IP 179.191.108.56 prioritized over the Domain Controller (10.2.17.2)?

A. It shows persistent beaconing behavior to an external, non-business IP.
B. It has a higher total byte count than the DC.
C. It's the only IP using the TCP protocol in the entire pcap.
D. It's located on the same subnet as the victim machine.

10. To investigate the specific port used by this suspicious IP, which query would we use to isolate its traffic?

A. `tcp.port == 443`
B. `addr.ip == 179.191.108.58`
C. `ip.addr == 179.191.108.58`
D. `ip.address == 179.191.108.58`

11. After isolating the IP, we see it communicating on port 449. Why is this significant in a malware investigation?

A. It's the default port for secure web browsing (HTTPS)
B. It indicates the host is attempting to use an encrypted VPN.
C. It proves that the attacker has successfully gained Domain Admin privileges.
D. It's a known non-standard port frequently used for Trickbot C2 communication.

12. When we want to inspect the TLS handshake details for this suspicious C2 connection, which query would we use?

A. `ip.addr == 179.191.108.58 && tls.handshake.type == 1`
B. `ip.addr == 179.191.108.58 && tls.type == clien_hello`
C. `tls_handshake_type == 1 && src.ip == 179.191.108.58`
D. `ip.address == 179.191.108.58 && ssl.handshake`

13. Using the TLS query, we see a variety of different Server Names (SNI). What does this suggest?

A. The user is visiting 50 different websites at the exact same time.
B. The DC is trying to load balance the external connection.
C. The malware is using SNI Spoofing to blend in with legitimate traffic.
D. The victim's browser cache is being cleared automatically by the system.

Phase Four: Persistence & Exfiltration

14. To check if the malware is attempting to steal stolen data (such as system info or cookies), which query would we use to find HTTP POST requests containing potential exfiltration?

A. `http.post.exfil == true`
B. `request.type == POST_DATA`
C. `ip.proto == HTTP && method == POST`
D. `http.request.method == "POST"`

15. In this pcap, we see POST requests to URIs ending in /81, /83, or /90. What do these numerical suffixes indicate in a rob13 infection?

A. Data Categorization: Each number identifies the type of stolen data (such as /81 for passwords).
B. Success Codes: They show the upload was 81% complete.
C. Encryption Keys: They represent the bit-length of the encryption used.
D. Server IDs: They represent which of the 90 different C2 servers is receiving the data.